

The High-Redshift Accretion Atlas

Project Status Through v4.0.1

Project status manuscript

23 August 2026

Abstract

JWST has identified accreting black-hole candidates at epochs where the available growth time is short and inferred physical quantities remain method-dependent. This project develops a reproducible, source-tracked, uncertainty-aware atlas that ranks observational leverage under explicit seed and accretion assumptions. Project v1 standardized 23 JADES broad-line AGN measurements at $z \geq 4$; v2 added point and Monte Carlo rankings plus figure prototypes; v3 added 37 Taylor et al. CEERS/RUBIES measurements; and v4 adds 20 Matthee et al. EIGER/FRESCO and 16 Lin et al. ASPIRE measurements. The corrected v4 release has 96 measurements representing 94 physical objects, explicit reviewed identity decisions, separate detection-confidence and mass-reliability semantics, and one-at-a-time sensitivity tests for both multiply measured objects. Maintenance release v4.0.1 pins the software environment, adds continuous integration and a hash manifest, hardens future identity allocation, and records source-specific virial calibrations without changing a mass, rank, or scientific scenario. Under the baseline $z_{\text{seed}} = 30$, $\epsilon = 0.1$, and no-merger reference, rankings express observational growth pressure and do not prove a unique seed or accretion channel. Next steps are final-manuscript consolidation, method-aware object classes, and duty-cycle diagnostics.

1 Scientific objective

The motivating question is simple: *how can the reported black holes become so massive so early?* The answer depends jointly on the initial seed mass, formation redshift, lifetime-average accretion rate, radiative efficiency, merger history, and the reliability of the inferred black-hole mass. Across the literature, these quantities are reported using different methods and assumptions. Consequently, the main project goal is not to draw one preferred growth track, but to construct an observational triage framework that separates:

- objects that remain growth-challenging under reasonable uncertainty and systematic tests;
- objects whose interpretation changes substantially with the inferred mass or growth assumptions; and
- measurements whose improvement would most change the physical conclusion.

The working thesis is that a standardized, provenance-aware atlas can identify a robust set of high-leverage objects for follow-up and detailed seed/accretion modeling. Formation scenarios are therefore treated as diagnostic coordinate systems, not as proof of a unique formation pathway.

2 Chronological development through v4

2.1 Stage 1: a controlled observational sample

The project begins with one relatively homogeneous class from one source paper: the JADES broad-line AGN catalogue of Juodžbalis et al. (2026) [2]. The source registry records the paper, tables used, selection, survey, and inference methods. The raw table contains 34 measurements; the reproducible standardization pass applies the $z \geq 4$ selection and produces 23 measurement rows spanning $z = 4.133\text{--}8.913$. The current quality classification contains 18 robust and 5 tentative measurements. Four high-redshift candidates—GS-20057765, GS-20030333, GS-164055, and GN-4685—use single-epoch broad- $H\beta$ virial masses and are explicitly classified as stack-supported tentative detections because their individual broad-component detections are not formally significant. Source-adopted CIGALE host masses are retained when BEAGLE is unavailable or the host is significantly extended; this supplies host masses for GS-200679, GS-20030333, and GS-164055 that were previously omitted.

The processed catalogue preserves coordinates, redshift provenance, black-hole mass and asymmetric uncertainty, host stellar mass, bolometric luminosity, reported Eddington ratio, inference methods, source-table provenance, notes, interpretation tags, a controlled detection-evidence category, derived quality flags, and explicit optional-field missingness. It also compares each reported Eddington ratio with the value implied by the tabulated black-hole mass and bolometric luminosity. Missing host or lensing information is retained and flagged rather than silently imputed.

Table 1: Catalogue chronology and current output scale.

Product	Current size
Raw source measurements	34
Processed measurements at $z \geq 4$	23
Robust / tentative measurements	18 / 5
Taylor Table 1 measurements (all redshifts)	63
Taylor measurements / objects at $z \geq 4$	37 / 36
v3 measurements / physical objects at $z \geq 4$	60 / 59
Matthee / ASPIRE source measurements	20 / 16
v4 measurements / physical objects at $z \geq 4$	96 / 94
Complete scenario-evaluation rows	8,832
Required- f_{Edd} rows	2,208
Required-seed rows	2,208
Per-object spin/merger parameter maps	138
Per-object seed-redshift maps	23

2.2 Stage 2: a reproducible physical model

The growth model follows the exponential form used by Dayal (2024) [1],

$$M_{\text{BH}}(t_{\text{obs}}) = M_{\text{seed}} \exp \left[f_{\text{Edd}} \left(\frac{1 - \epsilon}{\epsilon} \right) \frac{\Delta t}{t_{\text{Edd}}} \right] B_{\text{merge}}, \quad (1)$$

where $t_{\text{Edd}} \simeq 0.45$ Gyr and $\Delta t = t(z_{\text{obs}}) - t(z_{\text{seed}})$. Cosmic time is evaluated with a flat Planck-style cosmology using $H_0 = 67.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$, and $\Omega_\Lambda = 0.685$.

The model is used in three directions: predicting a final mass, solving for the required lifetime-average f_{Edd} at fixed seed mass, and solving for the required seed mass at fixed accretion history. Implemented

sensitivity dimensions include seed redshift, thin-disk radiative efficiency, Kerr spin, an illustrative photon-trapping efficiency above Eddington, and a multiplicative merger boost. Importantly, $B_{\text{merge}} = 2$ adds $\log_{10} 2 = 0.301$ dex to the mass; it does not double the exponential accretion rate.

2.3 Stage 3 (v2): from scenario grids to observational triage

The first evaluation stage generated growth tracks, fixed-scenario compatibility summaries, and per-object maps. These products show where each measurement lies across seed mass, accretion, efficiency, merger, and formation-redshift space. They also revealed an important interpretive distinction: a fixed scenario can miss an object through *undergrowth* or *overgrowth*. Only the former directly signals insufficient growth; the latter can be resolved by less accretion, a smaller seed, or later formation.

The project therefore moved from generic compatibility scores to required-parameter rankings. The baseline ranking uses $z_{\text{seed}} = 30$, $\epsilon = 0.1$, and $B_{\text{merge}} = 1$, and keeps three concepts separate:

1. **Physical pressure:** required f_{Edd} for fixed seeds and required seed mass for fixed accretion histories.
2. **Measurement evidence:** detection confidence is recorded separately from line-model and mass-measurement reliability; missing diagnostics are availability metadata rather than automatic penalties.
3. **Follow-up value:** the potential for a better measurement to change the physical ranking.

This produces a measurement-level table rather than a single opaque score. Host-mass tension, growth pressure, confidence, and follow-up priority remain independently inspectable.

2.4 Stage 4: uncertainty propagation

Point estimates are insufficient near physically meaningful thresholds. The uncertainty layer draws 10,000 samples per measurement from a split-normal approximation to the reported asymmetric $\log_{10} M_{\text{BH}}$ uncertainty. For every draw, the pipeline recomputes the required f_{Edd} and required seed mass. It reports the 5th, 16th, 50th, 84th, and 95th percentiles, together with threshold probabilities.

Three systematic mass scenarios are evaluated separately:

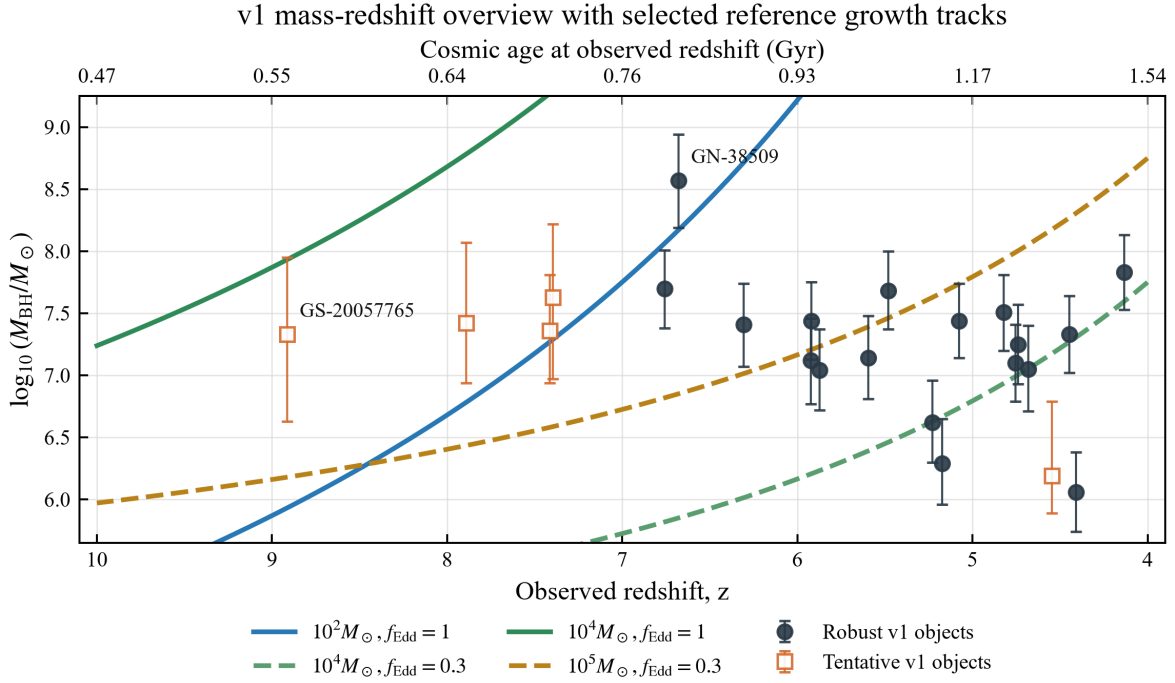
$$\text{baseline}, \quad \log_{10} M_{\text{BH}} - 0.3 \text{ dex}, \quad \log_{10} M_{\text{BH}} + 0.3 \text{ dex}. \quad (2)$$

The statistical mass sampling is never blended across these systematic scenarios. This separation makes it possible to distinguish reported statistical uncertainty from an assumed method-scale shift.

2.5 Stage 5 (v3): source expansion and physical-object identity

The v3 release adds the complete 63-row Table 1 extraction from Taylor et al. (2025) [3] as a source-specific raw table, then applies $z \geq 4$ in the processing layer. The retained Taylor subset has 37 measurements of 36 physical objects, including 17 sources marked as little red dots (LRDs). LRD is stored as a phenotype; the object class remains broad-line AGN. The combined release therefore contains 60 measurements of 59 physical objects.

Every Taylor row preserves coordinates, redshift, total/narrow/broad $\text{H}\alpha$ fluxes and uncertainties, broad-line FWHM, formal virial-mass uncertainty, LRD and absorption-fit markers, field/program, and source caveats. The Reines et al. (2013) $\text{H}\alpha$ single-epoch calibration is tagged precisely, and its approximate ± 0.5 dex virial systematic is evaluated separately from formal errors and the common ± 0.3 dex comparison. Host mass, bolometric luminosity, and Eddington ratio remain unavailable rather than inferred from absent Table 1 values.



Main-text prototype: points are source-reported v1 black-hole masses; curves are reference diagnostics, not fitted histories.
 All curves assume $z_{\text{seed}} = 30$, $\epsilon = 0.1$, and no merger boost; rankings are used for object triage.

Figure 1: The v1 JADES broad-line AGN sample in black-hole-mass-redshift space. The curves are selected reference growth histories, not fits to the observations. This overview establishes the timing problem before the model is inverted into required- f_{Edd} and required-seed diagnostics.

The measurement view retains both CEERS-2782 and RUBIES-EGS-50052. A separate crossmatch maps them to one stable physical-object ID; the higher-S/N RUBIES observation is the documented default for object-level ranking. No measurement is deleted or averaged. Summary tables are stratified by source, survey/field, LRD phenotype, and catalogue view, and mixed-selection rows are marked descriptive rather than demographic.

2.6 Stage 6 (v4): generalized identity and complementary WFSS samples

The v4 release adds all 20 Matthee et al. EIGER/FRESCO broad-H α emitters [4] and all 16 Lin et al. ASPIRE emitters [5] from authoritative arXiv source archives. Both papers use the Reines et al. (2013) H α single-epoch estimator and state a 0.5 dex calibration or intrinsic uncertainty. Native line luminosities, widths, photometry, continuum slopes, formal errors, LRD labels, absorption-fit flags, and row caveats are retained. Source-reported host mass and Eddington ratio remain missing; H α -derived bolometric luminosities are retained where published.

The identity layer performs deterministic coordinate/redshift candidate searches against both the prior release and other newly added sources, refuses ambiguous matches, and requires an explicit accepted/rejected decision in a reviewed override registry. Review confirms one new cross-paper identity: Matthee GOODS-S-13971 is JADES GS-204851 at 0.02 arcsec separation and $\Delta z = 0.001$. The prior-release JADES measurement remains the default for longitudinal reproducibility, while the Matthee row remains independently rankable. Together with the Taylor pair, v4 contains 96 measurements but 94 physical objects.

The v4 science workflow preserves the baseline assumptions and reported asymmetric errors, keeps the common ± 0.3 dex comparisons, and adds separately named ± 0.5 dex Reines-calibration scenarios for Taylor, Matthee, and ASPIRE. Detection confidence is now distinct from mass/line-model reliability, so robust broad-line detections with absorption or contamination caveats receive caveated follow-up categories without being relabelled as tentative detections. Object-level LRD unions retain supporting measurement and source provenance. Overall summaries explicitly mix unlike selection functions and are descriptive only.

The v4.0.1 method registry also resolves a previously implicit JADES distinction. The 19 H α measurements use the Reines & Volonteri (2015) relation and the source paper states a 0.3 dex calibration uncertainty. The four tentative H β measurements use Vestergaard & Peterson (2006); because the JADES paper does not assign that estimator a numeric systematic, the registry leaves it unavailable and retains the paper’s $A_V = 0$ underestimation caveat. These metadata are not folded into the reported statistical errors and do not retroactively alter the frozen v4 scenarios.

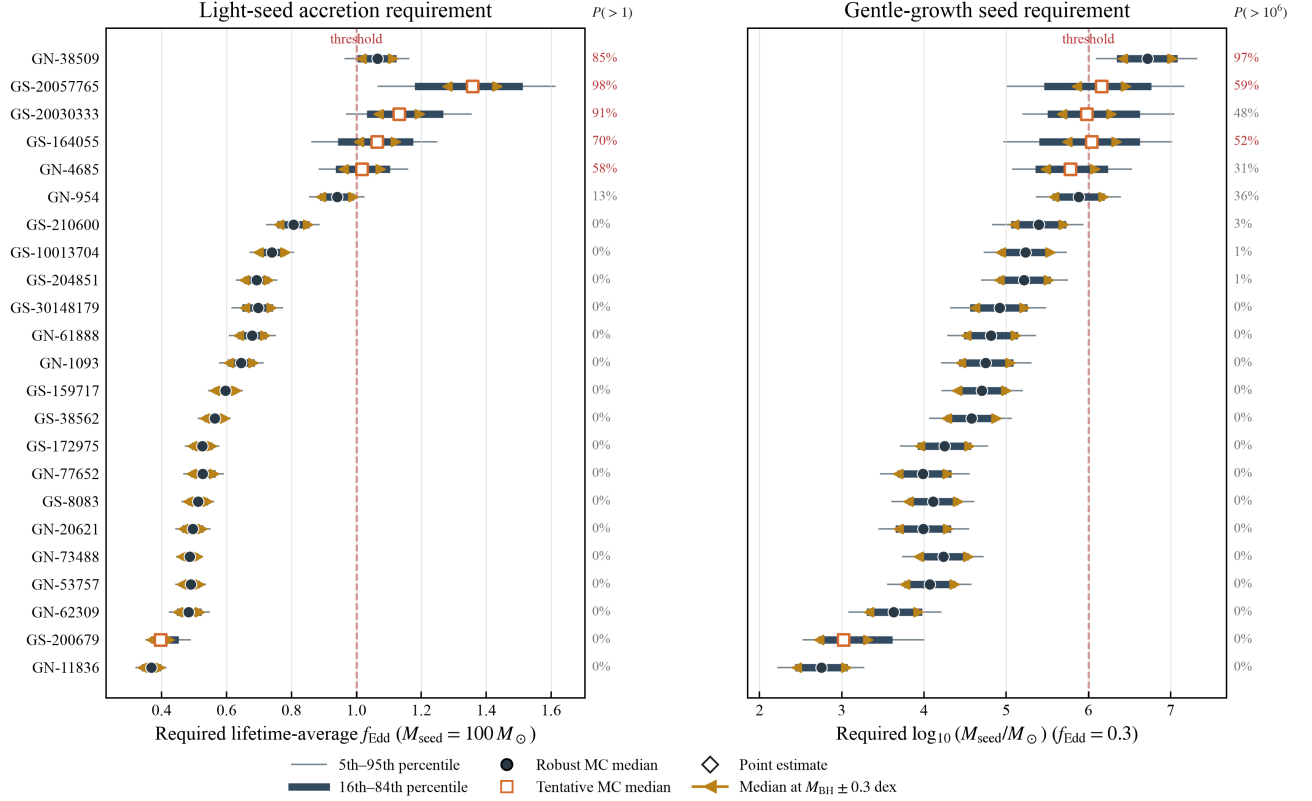
3 Current scientific findings

The v2 v1-catalogue analysis identifies a stable high-leverage group, but the uncertainty analysis changes how strongly individual objects should be described. The v3 object-level ranking retains GN-38509 and GS-20057765 in the first two positions and adds RUBIES-EGS-49140 in third; the Taylor object is flagged for possible outflow contribution or an alternative non-AGN broad-line interpretation. Table 2 retains the detailed v1 probabilities that underlie the first comparison.

GN-38509 is currently the clearest robust result: it remains demanding in both light-seed and gentle-growth diagnostics after propagating the reported mass uncertainty. GS-20057765 is more extreme in the light-seed calculation but is tentative and has a broader required-seed distribution. It is therefore a high-priority follow-up target rather than an equally secure theoretical constraint. GN-954 illustrates why probability statements matter: its point estimate is near a pressure boundary, but its baseline probability of requiring super-Eddington lifetime-average growth from a $100 M_\odot$ seed is only 0.13.

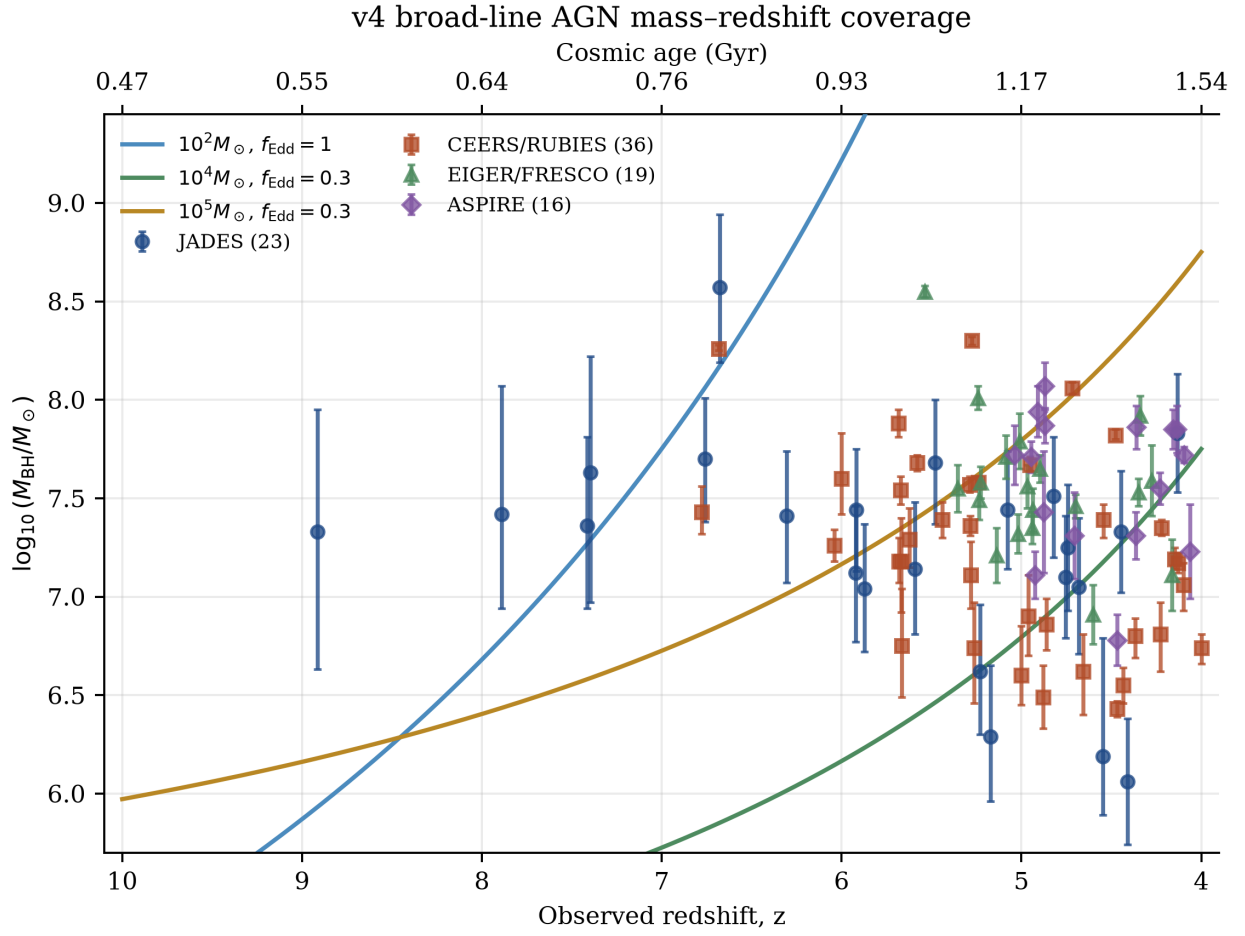
The current reported Eddington ratios are instantaneous observational estimates, whereas the model-

Uncertainty-aware growth-pressure ranking



Baseline Monte Carlo intervals propagate reported asymmetric M_{BH} errors; percentages give baseline threshold probabilities.
 Orange endpoints are separate $M_{\text{BH}} - 0.3$ and $+0.3$ dex systematic medians; all panels assume $z_{\text{seed}} = 30$, $e = 0.1$, and no merger boost.

Figure 2: Uncertainty-aware growth-pressure ranking. Thin and thick intervals show the baseline 5th–95th and 16th–84th percentiles. Circles and open squares denote robust and tentative Monte Carlo medians; diamonds are the original point estimates. Orange endpoints show the separate $M_{\text{BH}} \pm 0.3$ dex systematic medians. Red dashed lines mark $f_{\text{Edd}} = 1$ for a $100 M_{\odot}$ seed (left) and $M_{\text{seed}} = 10^6 M_{\odot}$ for $f_{\text{Edd}} = 0.3$ (right). Percentages give baseline probabilities of crossing the corresponding threshold.



Tracks assume $z_{\text{seed}} = 30$, $\varepsilon = 0.1$, merger boost 1; they are diagnostics, not fitted histories.

Figure 3: v4 physical-object mass-redshift coverage, stratified across the four source selections. Reference tracks use the repository baseline and are diagnostic rather than fitted histories.

Table 2: Highest-leverage v1 measurements under the baseline uncertainty model.

Object	Quality	$P(f_{\text{Edd,req}}^{100} > 1)$	$P(M_{\text{seed,req}}^{0.3} > 10^6 M_{\odot})$	Status
GN-38509	robust	0.85	0.97	strongest robust pressure case
GS-20057765	tentative	0.98	0.59	strongest tentative follow-up target
GS-20030333	tentative	0.91	0.48	high leverage; elevated BH/host ratio
GS-164055	tentative	0.70	0.52	high leverage; extreme BH/host ratio
GN-4685	tentative	0.58	0.31	threshold and systematics sensitive
GN-954	robust	0.13	0.36	comparison/systematics object

derived f_{Edd} values are lifetime averages. Their difference is scientifically important but should not be interpreted as a contradiction until duty cycles and burst histories are modeled. A separate source-table cross-check identifies GN-11836 as internally inconsistent: its published $\log M_{\text{BH}} = 6.06$ and $\log L_{\text{bol}} = 44.11$ imply $\lambda_{\text{Edd}} \simeq 0.89$, while the table reports 0.11. All published values are retained pending clarification, and the measurement is downgraded to medium confidence for Eddington-ratio comparisons.

The corrected v4 point ranking keeps GN-38509 first and GS-20057765 second. RUBIES-EGS-49140 and GOODS-N-9771 rank third and fourth in physical growth pressure, but both are explicitly caveated for mass/line-model interpretation while retaining high detection confidence. Substituting CEERS-2782 for the default RUBIES-EGS-50052 measurement lowers the adopted mass by 0.42 dex and moves that physical object from growth-pressure rank 28 to 46; substituting Matthee GOODS-S-13971 for JADES GS-204851 lowers the mass by 0.19 dex and moves the object from rank 17 to 22. These are sensitivity results, not changes to release defaults.

4 Release map and reproducibility

The project versions are chronological milestones: v1 is the pilot JADES catalogue and baseline evaluation, v2 is ranking/uncertainty/figure analysis on that frozen catalogue, v3 is the combined JADES + Taylor catalogue, and v4 generalizes identity while adding Matthee and ASPIRE. Source-paper versions, including arXiv suffixes, are independent provenance metadata. The canonical dependency map is:

1. `data/sources.md` and `data/raw/v1_raw.csv`: source provenance and extracted measurements.
2. `docs/catalogue-schema.md`: canonical fields, validation rules, and missing-value policy.
3. `src/standardize_data.py` and `scripts/process_data.py`: raw-to-processed catalogue construction.
4. `src/models.py`: cosmology and black-hole growth calculations.
5. `src/scoring.py` and `scripts/v1_evaluate.ipynb`: scenario evaluation and exploratory compatibility products.
6. `scripts/generate_v2_rankings.py`: v2 point-estimate physical-pressure and follow-up rankings from v1.
7. `scripts/generate_v2_uncertainty_rankings.py`: v2 deterministic Monte Carlo summaries and probability-based rankings.
8. `scripts/generate_v2_final_figures.py`: v2 publication-style figures driven by the v1-catalogue result tables.
9. `scripts/process_v3_blagh.py` and `src/v3_catalogue.py`: v3 source combination, crossmatch, and object view.

10. `scripts/generate_v3_blagh_science.py` and `src/v3_science.py`: v3 measurement/object evaluation, rankings, uncertainty, and summaries.
11. `scripts/generate_v3_final_figures.py`: frozen-v3 comparison figures.
12. `scripts/process_v4_blagh.py`, `src/identity.py`, and `src/v4_catalogue.py`: v4 ingestion, pairwise candidate generation, explicit reviewed decisions, aliases, and measurement/object views.
13. `scripts/generate_v4_blagh_science.py` and `src/v4_science.py`: v4 point and uncertainty products, confidence/reliability separation, phenotype attribution, and duplicate-measurement sensitivity.
14. `requirements-lock.txt`, `releases/v4.0.1-manifest.json`, and `scripts/verify_v4_release.py`: exact dependencies, frozen hashes, and write-free in-memory reproduction.
15. `data/mass_method_registry.csv`: reviewed source/method calibrations and explicitly unavailable numeric systematics.
16. `scripts/generate_v4_final_figures.py`: the five corrected-v4 figures.
17. `tests/test_v1_v2_pipeline.py` and the `test_v3_*/test_v4_*` modules: model, schema, release, provenance, ranking, uncertainty, and regression checks.

Continuous integration installs the pinned Python 3.12 environment, runs the complete regression suite, verifies every manifest hash, rebuilds all 18 frozen v4 catalogue/science CSVs in memory at 10,000 draws with seed 20260808, and requires the checkout to remain clean. These checks protect implementation behavior, release byte anchors, source counts and identity links, explicit review decisions, confidence semantics, systematic separation, measurement sensitivity, and current numerical rankings; they do not establish that the astrophysical assumptions are uniquely correct.

5 Limitations

The present atlas remains a selected-sample triage framework, and its conclusions should be read within these boundaries:

- v4 contains JADES, CEERS/RUBIES, EIGER/FRESCO, and ASPIRE selections, not a single complete population sample, so pooled rows cannot support demographic inference;
- the main growth diagnostics assume constant lifetime-average accretion rather than explicit duty cycles or bursts;
- the split-normal mass model approximates reported asymmetric uncertainties but is not a reconstruction of each source’s full posterior;
- the ± 0.3 dex cases are transparent comparison tests and the Taylor/Matthee/ASPIRE ± 0.5 dex cases are calibration sensitivities, not reconstructed systematic posteriors;
- the four stack-supported tentative broad- $H\beta$ candidates require individual spectroscopic confirmation before they can serve as secure theoretical constraints;
- the GN-11836 source-table Eddington-ratio inconsistency must be clarified before that ratio is used quantitatively;
- Taylor Table 1 does not publish host mass, bolometric luminosity, or Eddington ratio, so those diagnostics are unavailable rather than negative evidence; and
- deterministic physical-object linkage is explicit, but fixed thresholds and manual review are not a substitute for probabilistic crossmatching in crowded or low-precision samples.

These limitations motivate the next development stages rather than invalidating the release-stratified comparison framework.

6 Research roadmap

The next work should proceed in the following order.

1. Consolidate and audit the final manuscript

Integrate the completed corrected-v4 figures and tables, verify every numeric claim and citation, and decide which source-stratified products belong in the main text or appendix.

2. Add the next measurement-version layers

Attach Harikane and earlier discovery-paper measurements to stable v4 objects, improving alias coverage while retaining every literature row and rejecting ambiguous automatic matches.

3. Add object classes without erasing method differences

Extend the atlas to little red dots, X-ray candidates, lensed candidates, high-ionization-line candidates, and classical luminous quasars. Each class will retain distinct selection and mass-method metadata so that heterogeneous measurements are not treated as interchangeable.

4. Improve the uncertainty model

Propagate host-mass and lensing uncertainties where available, add method-specific black-hole-mass systematic scenarios, and use fuller posterior information when a source provides it. The output should continue to emphasize intervals and threshold probabilities rather than categorical point-estimate labels.

5. Replace constant-average growth with history diagnostics

Introduce duty-cycle and bursty-accretion models, compute the required duty cycle under fixed burst assumptions, and compare the reported current Eddington ratio with the required lifetime-average value without treating them as the same observable.

6. Convert the ranking into an observing strategy

Produce a follow-up matrix specifying whether each object’s interpretation is most sensitive to deeper spectroscopy, improved virial or dynamical mass constraints, host/AGN decomposition, X-ray observations, lensing reconstruction, or repeat measurements.

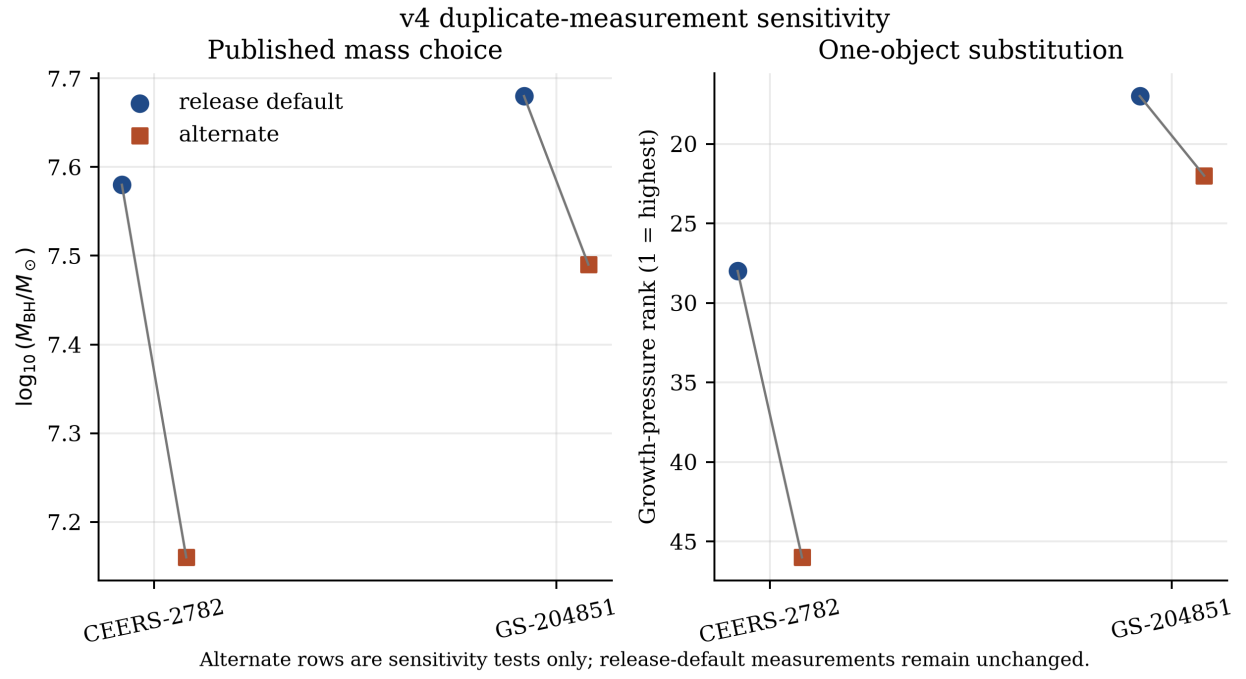
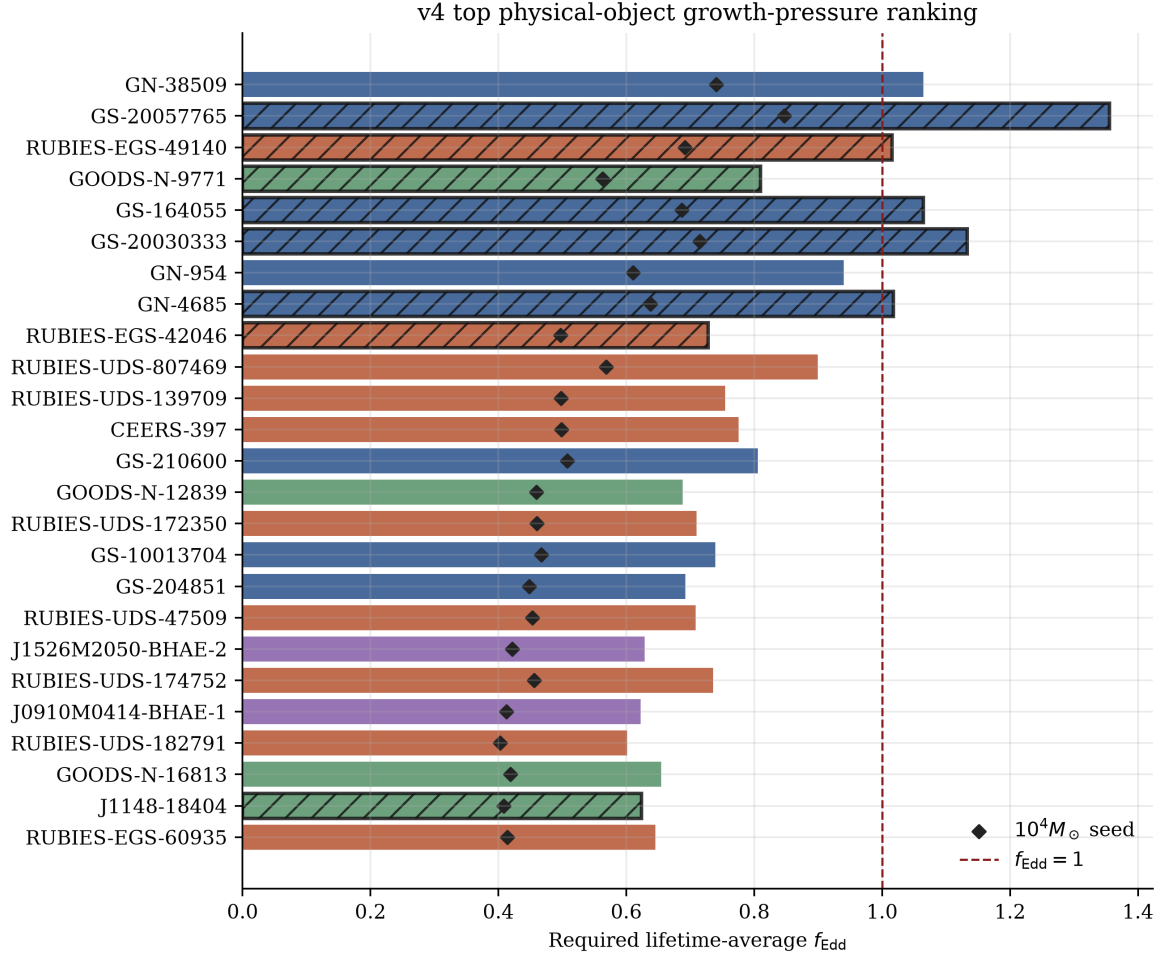


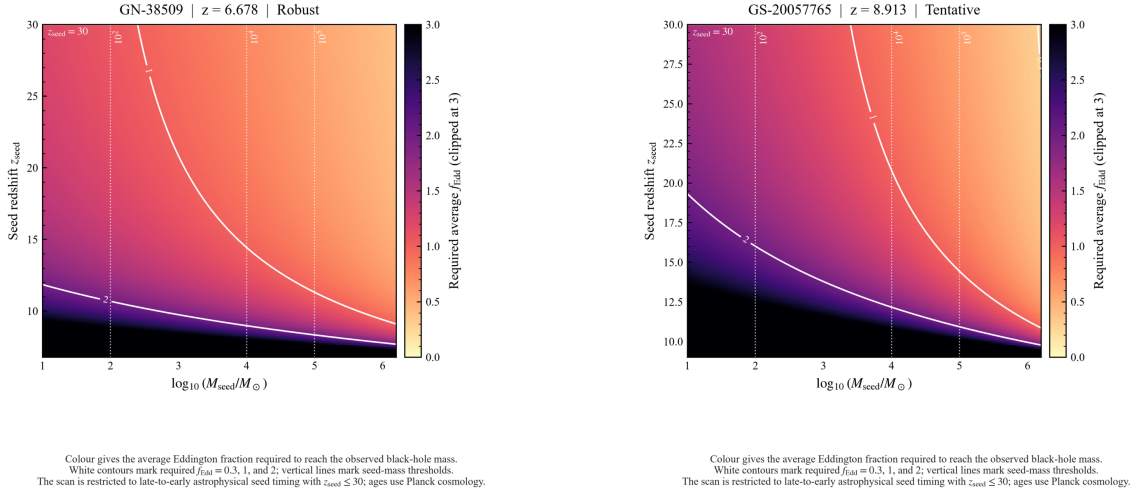
Figure 4: One-at-a-time sensitivity to the nondefault literature measurement for each multiply measured physical object. The exercise recomputes the full 94-object ranking while preserving all published rows and the documented release defaults.



Hatched bars flag mass/line-model caveats; robust detection confidence is recorded separately.

Figure 5: Corrected-v4 physical-object growth-pressure ranking. Bar colour records the preferred-measurement source, while hatching flags mass/line-model caveats independently of broad-line detection confidence. Each physical object appears once.

Spotlight seed-timing maps for the two strongest v1 leverage cases



Existing seed-redshift maps are reused as main-text spotlight prototypes. GN-38509: growth-pressure rank 1, $f_{\text{Edd}}(100M_{\odot}) = 1.06$; GS-20057765: growth-pressure rank 2, $f_{\text{Edd}}(100M_{\odot}) = 1.36$. Panels diagnose assumption sensitivity across seed mass and seed redshift; they do not select a unique formation path.

Figure 6: Seed-redshift spotlight maps for GN-38509 and GS-20057765. These maps show how the required lifetime-average accretion rate changes with seed mass and formation time. The full object gallery is retained for supplementary use; selected panels connect the sample ranking to object-level physical interpretation.

7 Conclusion

The completed v1–v4 work demonstrates the atlas concept: provenance-aware ingestion, validated standardization, explicit reviewed measurement/object identity, transparent physical modeling, point and Monte Carlo rankings, selection-aware summaries, duplicate-measurement sensitivity, release-specific visualization, and automated regression checks. v4 broadens the lower-redshift end of the same-class BLAGN comparison while preserving every literature row and preventing double-counting. This is not evidence that one formation pathway is required. The next phase should finish the manuscript audit, then add carefully matched measurement-version layers, method-aware object classes, and more realistic accretion histories.

References

- [1] Dayal, P. 2024, *A&A*, 690, A182.
- [2] Juodžbalis, I. et al. 2026, “JADES: comprehensive census of broad-line AGN from Reionization to Cosmic Noon revealed by JWST,” *MNRAS*, 546, stag086, arXiv:2504.03551.
- [3] Taylor, A. J. et al. 2025, “Broad-Line AGN at $3.5 < z < 6$: The Black Hole Mass Function and a Connection with Little Red Dots,” *ApJ*, 986, 165, doi:10.3847/1538-4357/add15b.
- [4] Matthee, J. et al. 2024, “Little Red Dots: an abundant population of faint AGN at $z \sim 5$ revealed by the EIGER and FRESCO JWST surveys,” *ApJ*, 963, 129, doi:10.3847/1538-4357/ad2345.
- [5] Lin, X. et al. 2024, “ASPIRE: Broad-line AGN at $z = 4$ –5 revealed by JWST/NIRCam WFSS,” *ApJ*, 974, 147, doi:10.3847/1538-4357/ad6565.